StudentsCatalyst

Smart Bins, Smart People, and Smart Data with I.B.D Strategy

Regular Forum #3



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Executive Summary

Summarize Solution: Smart Bins, Smart People, and Smart Data with I.B.D Strategy

Campus waste **sorting is still weak**, and the system is not yet data-driven.

Re-POTE

Large campus activity creates high daily waste volume and makes manual sorting hard to manage.

Situation

The campus wants better waste sorting, but low awareness and weak monitoring still make the system ineffective.

High student activity means high waste volume and more pressure on the sorting system.

End-to-End Solution to tackle the problem

- *Infrastructure Enhancement* → Smart sorting and recycling points with sensors and clear sorting support
- *Behavior & Awareness Improvement* → Student reward system to make sorting easier and more consistent
- *Data & Monitoring System Optimization* → Real-time dashboard to track waste, participation, and performance

Key Question

How can the campus improve waste sorting through better student habits, Reduce plastic waste by recycling, continuous education, and an integrated monitoring system?

Answer



Expected Impact

Higher Sorting Accuracy

Improve sorting accuracy through clearer bins, smart support, and better student action

Real-Time Waste Monitoring

Track waste type, volume, and activity automatically through the dashboard

Data-Driven Sustainability

Turn campus recycling into a measurable system that can be improved over time

Build Students Behavior

Build daily sorting habits through rewards, engagement, and feedback

Empowering Campus Recycling

By connecting infrastructure, data monitoring, and community participation, Re-POTE creates an integrated waste management ecosystem for campuses.

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Campus Waste Landscape

Global

Emerging Report:

70Tons *Waste/day at campus*

80Kg *Waste/year per person*

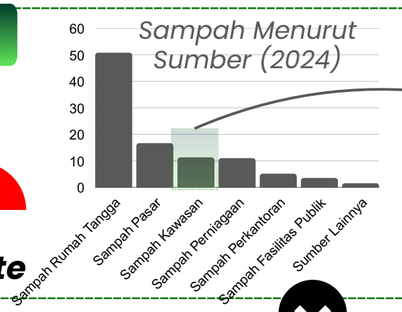
45% *of campus waste is plastic & paper*

Campus Waste

Sc: Business Waste

The high volume of waste reflects a culture that remains very linear (take, use, dispose of) and is not very sustainable

Indonesia



Activity Waste

12.15% *From Daily Activity Waste*

61% *Unmanaged Waste*

Sc: Detiknews, Goodstats

Current Trend:

- ESG & Sustainability Reporting
- Growing Waste Management Innovation
- Food delivery dan packaging

Re-POTE Sustainability Overview

Re-POTE

Indonesia Emas 2045 to Support Waste Management Program

Re-POTE Target:

-99% *Target **waste diversion** (2030)*

-13.5% *Target **waste reduction** (2030)*

Campus Current Conditions:

>100K Student → **>100Ton** *Campus waste/day (all area, assumption 0.5kg/student)*

40% *Rate waste sorting accuracy*

Sc: Databox

Gaps driven with current condition

- **Low awareness** to segregation
- The infrastructure is insufficient to support waste segregation
- A one-time awareness campaign on waste

The resulting impact:

- Decline in Waste Management Efficiency
- Increase in the Volume of Residual Waste Sent to the Landfill
- Rising Transportation and Processing Costs

Key Question to Solve:

How does Re-POTE improve the effectiveness of waste sorting through behavioral change, ongoing education, and an integrated monitoring system?

Key Area to Analyze:

Internal: People, Process, Product dan Management

External: Competitor, Trend, dan PESTLE overview

Root Cause Answered:

A1 A2 B1

Solution 2 - Behavior & Engagement Improvement

Recycle & Reward Program to Increase Student Participation on Campus

Overview



Many students **care about sustainability**, but recycling is still **not a strong daily habit**.



Students **often choose the easiest** option, so **recyclable waste still ends up in the wrong bin.**

Key Solution

"Sort, Scan, Sustain"

Reward-Based Recycling to Build Daily Student Habits



Sort Waste at the Smart Bin



Scan and Earn EcoPoints



Onsite Campaign



Student Engagement

How does it work?

1 Sort Waste at the Smart Bin

Students throw bottles, paper, and other recyclable waste into the correct bin.



Make recycling easy in daily campus life

2 Scan and Earn EcoPoints

Students scan the QR code or tap their student card after recycling.



The system records the activity and gives digital points automatically.

Every correct action gives students EcoPoints

3 Redeem Rewards and Join Challenges

- Students can **exchange points for rewards** such as vouchers, tumblers, tote bags, lunch boxes, and other reusable items.
- They can also **join faculty challenges** and campus leaderboard activities.

Rewards make recycling more fun and more consistent

4 Student Engagement Metrics

Recycling participation becomes a measurable student activity on campus

- Student Participation Rate (%)
- Repeat Recycling Users & Points Redeemed
- Recyclable Waste Collected

Impact

-20%
Waste reduced each students

70% → 90%
Waste sorting accuracy

(PwC, 2025 ; AHM Report, 2025)

Root Cause Answered:

B2 B1 C2 C1

Solution 3 - Data & Monitoring System Optimization

Monitoring Dashboard to improve process input of waste

Objective

To build a **monitoring system that connects smart bin data and student activity,** so the campus **can track waste better and improve recycling performance.**

Need a smarter way to monitor campus recycling

Key Features

- Smart Bin Integration
- Real-Time Data Sync
- Dashboard Monitoring
- Reporting & Analytics
- User Roles & Access
- Gamification & Rewards

Expected Sustainability Impact

Turns campus recycling into a measurable system that can be tracked, improved, and scaled.

Quick Wins

Smart Bin Integration



The dashboard connects directly with smart recycling points, so waste data can be recorded automatically and viewed in one system

Smart Bin Integration

Connects bin activity to the dashboard

Real-Time Data Sync

Updates waste and usage data automatically

Student Points & Leaderboard



Real-Time Data Sync

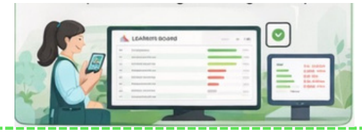
Updates waste and usage data automaticall

Dashboard Monitoring

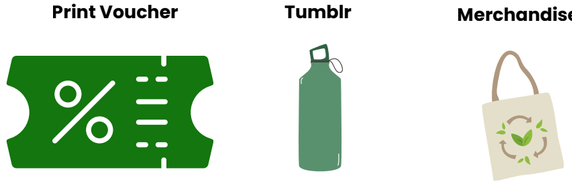
Shows key campus recycling performance in one place

Participation Tracking

Students collect EcoPoints, track their activity, and join faculty or campus challenges through the system.



Reward



Strategic Partners for Campus Waste Infrastructure

Key partners that support implementation, collection, monitoring, and public engagement to build an effective campus recycling system.

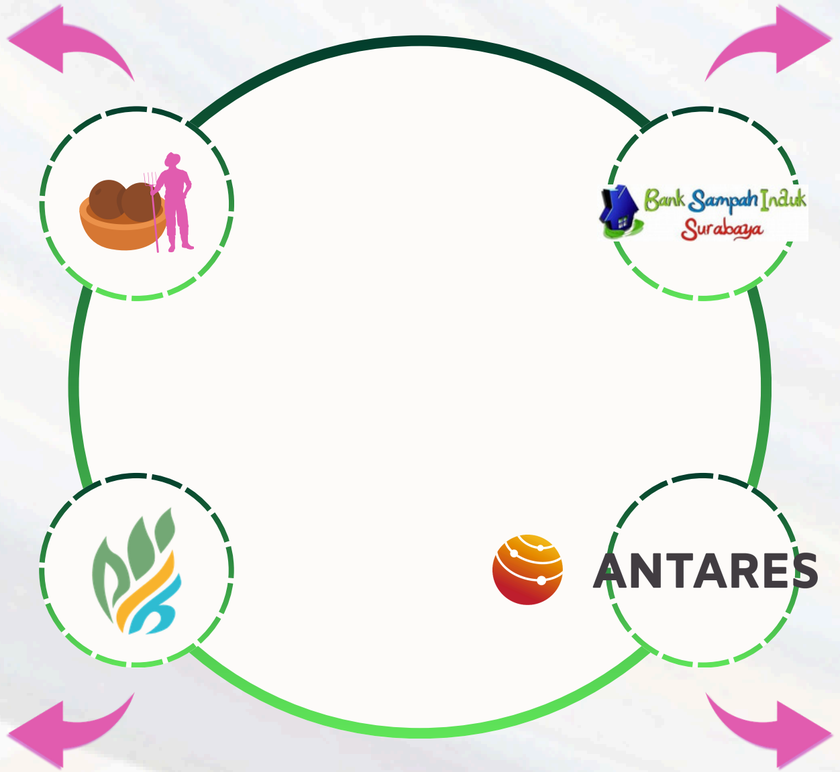
Priority Partners

Campus in Surabaya

A strong testbed for installing recycling points, running student participation programs, and **validating the system in a real campus environment.**

Bank Sampah Induk Surabaya

Can handle the pickup and downstream processing of sorted recyclables from campus recycling points



Dinas Lingkungan Hidup

Can support environmental campaigns, public legitimacy, and alignment with local **waste-management programs.**

Telkom Antares

Can support smart bin sensors, real-time tracking, and dashboard **integration for waste data monitoring.**

Values Proposition

From Waste Handling to Intelligent Waste Management System

Operational Efficiency

- Reduce manual sorting & monitoring effort
- Optimize waste handling & collection process
- Lower operational and logistics cost

Behavioral Transformation

- Shift responsibility from “cleaning staff responsibility”→ individual accountability
- Create daily habit, not one-time campaign
- Reinforce behavior through continuous feedback loop

Sustainability

- Support Zero Waste to Landfill goals
- Enable ESG-ready reporting & audit compliance
- Strengthen corporate sustainability positioning



Data Visibility & Control

- Real-time tracking of waste volume & accuracy
- Department-level performance monitoring
- Enable faster, data-driven decision making

Impact

90%	-30–40%	Real-time
sorting accuracy	residual waste	tracking across departments

Implementation Plan

The implementation follows a disciplined, phased approach designed to secure Product-Market Fit (PMF) before scaling across the entire national network.

Phase 1: Quick Wins & Pilot Launch (Year 1)

- Targeted Infrastructure: Deploy "Smart Waste Bins" equipped with volume, moisture, and weight sensors in two pilot plants.
- System Foundation: Launch the initial dashboard to monitor waste volume, error rates, and automated pickup scheduling.
- Behavioral Kickoff: Initiate the "<2 kg waste per employee per day" awareness campaign and Phase 1 of gamification, featuring division-based leaderboards.

Phase 2: Ecosystem Expansion (Year 2)

- Full Rollout: Expand the hardware installation across all five major plants.
- Advanced Analytics: Upgrade the monitoring system to a real-time dashboard with predictive alerts.
- Strategic Partnerships: Establish formal collaborations with external recycling partners for downstream processing of plastic, paper, and metal.
- Digital Engagement: Roll out digital badges and "Division Champion" rewards to sustain student/employee participation.

Phase 3: Integration & AI Optimization (Year 3)

- Enterprise Integration: Sync the waste segregation data into existing ERP/Manufacturing systems (e.g., SAP/Oracle).
- Technical Upgrade: Fully implement AI Vision technology to push sorting accuracy to >= 95%
- Standardization: Finalize the Waste Management Framework as the standard operating procedure for the national network.

Impact Analysis

Re-POTE transforms waste handling from a manual burden into an intelligent, data-driven ecosystem.

Operational & Environmental Impact

- Sorting Accuracy: Expected to increase significantly, targeting 90% accuracy through smart sensor support.
- Waste Reduction: A projected 30-40% reduction in residual waste volume sent to landfills.
- Efficiency Gains: Reduces manual labor costs and optimizes logistics, leading to an 18% decrease in waste transport trips.
- Carbon Footprint: Estimated reduction of 8-12 tons of CO_2 per year.

Financial Feasibility

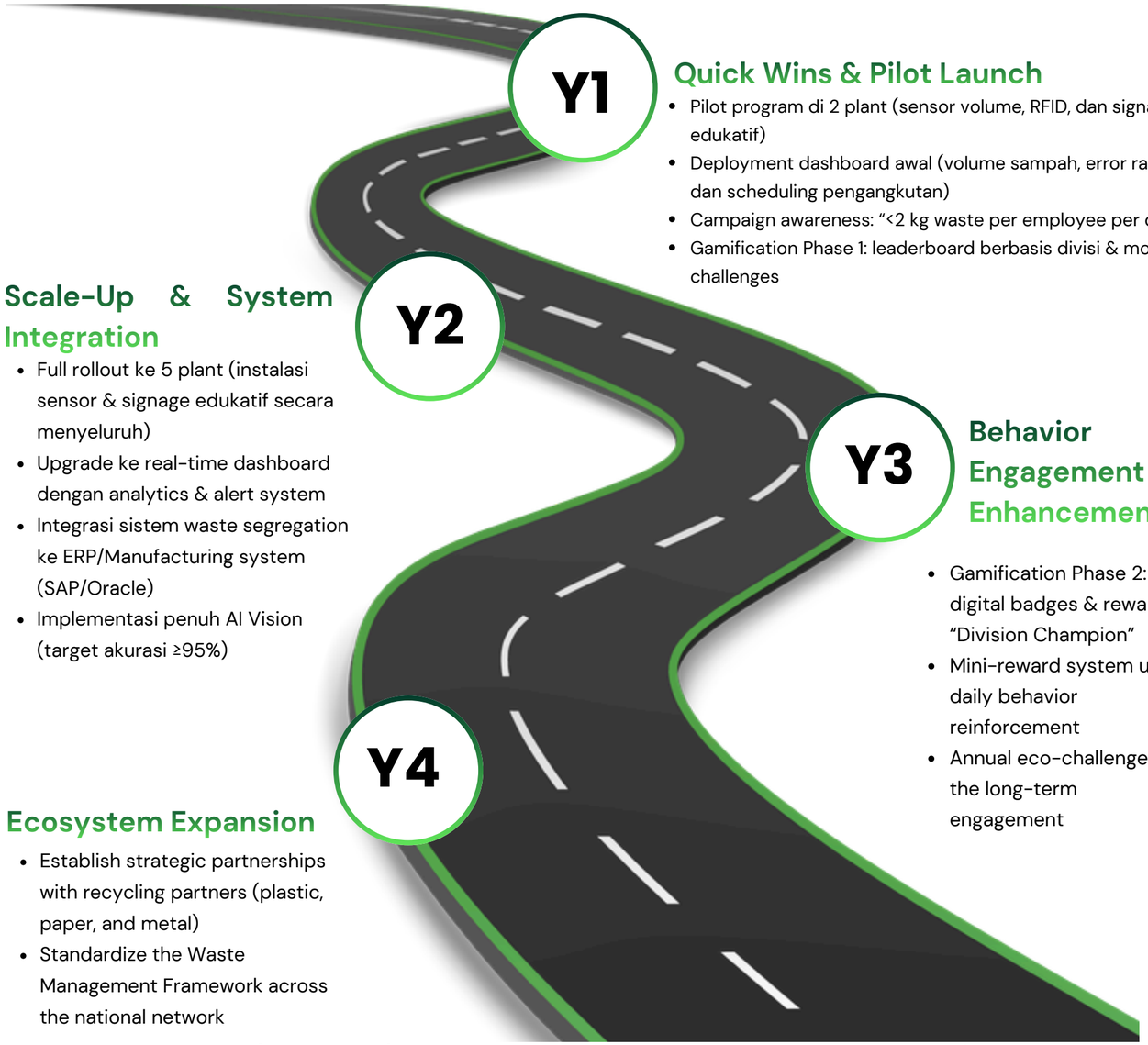
- ROI & IRR: A 10-year ROI of 27.46% and an Internal Rate of Return (IRR) between 28-32%.
- Payback Period: The system is expected to reach the break-even point in 3 years and 11 months.
- Benefit/Cost (B/C) Ratio: A healthy ratio of 1.21, indicating the benefits outweigh the costs.

Social & Behavioral Transformation

- Accountability Shift: Successfully moves the culture from "cleaning staff responsibility" to individual accountability.
- Sustained Engagement: Through gamification, employee/student participation is projected to grow from an initial 60% to 85% by Year 3.
- Data Visibility: Provides real-time, department-level performance monitoring, enabling faster, data-driven sustainability decisions.

Feasibility Study

Assessing the solution through ESG, Financial, and Operational Readiness lenses throughout the implementation timeline.



Quick Wins & Pilot Launch

- Pilot program di 2 plant (sensor volume, RFID, dan signage edukatif)
- Deployment dashboard awal (volume sampah, error rate, dan scheduling pengangkutan)
- Campaign awareness: "<2 kg waste per employee per day"
- Gamification Phase 1: leaderboard berbasis divisi & monthly challenges

Scale-Up & System Integration

- Full rollout ke 5 plant (instalasi sensor & signage edukatif secara menyeluruh)
- Upgrade ke real-time dashboard dengan analytics & alert system
- Integrasi sistem waste segregation ke ERP/Manufacturing system (SAP/Oracle)
- Implementasi penuh AI Vision (target akurasi ≥95%)

Behavior Engagement Enhancement

- Gamification Phase 2: digital badges & reward "Division Champion"
- Mini-reward system untuk daily behavior reinforcement
- Annual eco-challenge for the long-term engagement

Ecosystem Expansion

- Establish strategic partnerships with recycling partners (plastic, paper, and metal)
- Standardize the Waste Management Framework across the national network

How does it works?

B/C Ratio	ROI 10 Tahun
1,21	27,46%

Payback Period

3 Years, 11 Months

Environmental Social Governance (ESG) Menggunakan Boda Count Method

